
Organic Rankine Cycle Power Block

Control cycle and electrical simulation model
for WECC interconnection



Revision history

REV.	DATE	AUTHOR	DESCRIPTION
0.1	Sep. 4, 2026	Eugen Starschich	Initial document

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1 Electrical simulation

1.1 Scope and reference configuration

This section documents the control cycle of the Organic Rankine Cycle (ORC) power block and the electrical model derived from it for grid interconnection studies. The reference configuration is a **40 MW-class ORC unit, directly grid-connected through a synchronous generator** — the arrangement used by large ORC plants (Ormat and Turboden class, roughly 0.3–20 MW per unit and above), in which the expander drives a conventional four-pole synchronous machine at 1,500/1,800 rpm through a reduction gearbox.

Small ORC units below roughly 300 kW, and high-speed direct-drive machines, instead use a permanent-magnet generator with a rectifier and a grid-tie inverter; such a plant is modelled with the WECC generic inverter-based resource models (REGC_A / REEC_A / REPC_A) and is outside the scope of this section.

The target interconnection is the **Western Interconnection (WECC)**, which drives the model-acceptance and power system stabilizer requirements in Section 1.7.

1.2 Plant architecture and control cycle

Figure 1 shows the complete plant: the energy path from heat source to point of interconnection, the closed working-fluid loop, and the control and measurement layer wrapped around them.

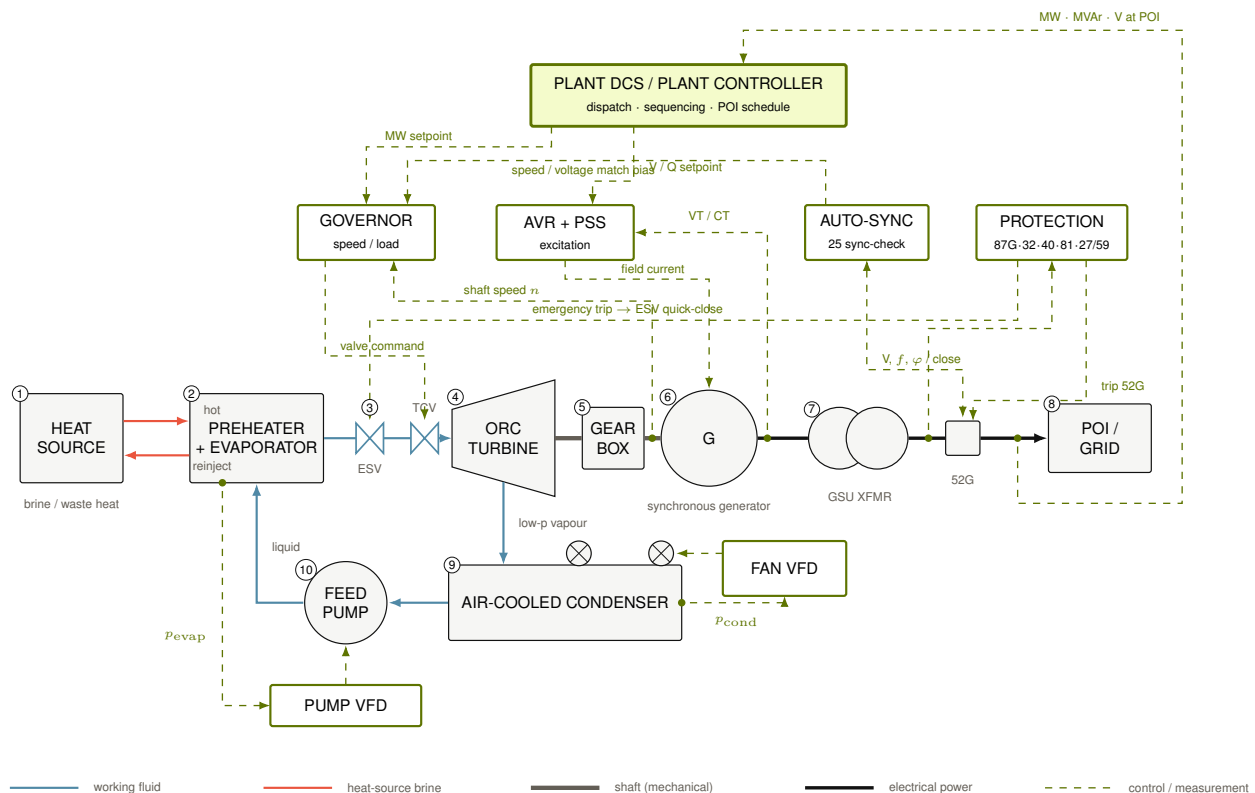


Figure 1: ORC plant control cycle. Energy flows left to right; the working fluid circulates evaporator → turbine → condenser → feed pump. Numbers key to Table 1. Setpoint links from the plant DCS to the pump and fan VFDs, and the recuperator common on many ORC cycles, are omitted for legibility; neither adds a control loop.

1.2.1 Physical parts of the energy path

#	Part	Function and relevance
1	Heat source loop	Geothermal brine or waste-heat stream with its circulation pumps. Sets the total heat available and is the effective fuel valve of the plant.
2	Preheater, evaporator	Heats and vaporises the organic working fluid. Key measurements: evaporator pressure and superheat.
3	ESV and TCV	Spring-closed emergency stop valve (tripped by protection) and the throttle valve modulated by the governor. The only fast handles on turbine power.
4	ORC turbine	Axial or radial-inflow expander converting vapour enthalpy to shaft torque. Fitted with speed pickups and an independent overspeed trip.
5	Gearbox	Reduces turbine speed to synchronous speed (1,500/1,800 rpm). Omitted on machines designed to drive the generator directly.
6	Synchronous generator	Four-pole machine with shaft-mounted brushless exciter. Field current from the AVR sets terminal voltage and reactive power. Modelled as GENROU.
7	GSU transformer	Steps generator voltage (typically 11–13.8 kV) up to interconnection voltage. Carries its own differential protection zone.
8	Generator breaker 52G, POI	Switching point between island and grid-connected operation, closed only on synchroniser permission. Revenue metering and the regulated POI quantities sit here.
9	Air-cooled condenser	Rejects cycle heat to ambient; VFD-driven fan motors set condensing pressure. A cooling-tower loop replaces it on water-cooled plants.
10	Working-fluid feed pump	VFD-driven pump closing the loop. Its speed is the principal handle on working-fluid mass flow, and therefore on plant output over minutes.

Table 1: Physical parts of the ORC energy path, keyed to Figure 1.

1.2.2 Control loops

Each loop pairs a sensor, a controller and an actuator around one of the parts above. The fast loops (milliseconds to seconds) reside in dedicated hardware — governor, AVR, protection — while the slow thermal loops (seconds to minutes) reside in the plant DCS. Only the fast loops survive into the dynamic model of Section 1.4.

1.3 Protection

Protection acts once and latches rather than regulating continuously. The generator protection relay — differential (87G), reverse power (32), loss of field (40), over/under-frequency (81) and over/under-voltage (27/59) — trips the 52G breaker and simultaneously commands the turbine trip system, which dumps hydraulic pressure and allows the spring-closed ESV to shut. The reverse-power element is of particular relevance to an ORC unit: left connected without working-fluid flow, the machine would motor the generator and drive the expander backwards.

An independent electronic overspeed system trips the ESV directly, without passing through the DCS, so that loss of the control system can never disable overspeed protection. The GSU transformer carries its own differential and overcurrent zone.

In the simulation model these functions appear as breaker events and relay settings, not as dynamics.

1.4 Positive-sequence dynamic model

In a positive-sequence dynamic model (PSS/E, PSLF, PowerFactory, ETAP) the whole plant of Figure 1 collapses to three interconnected blocks plus the network solution. The thermal loops — evaporator pressure,

Loop	Measured	Actuator	Regulated quantity
Speed / load	Shaft speed, MW output, turbine inlet pressure	TCV (3)	Speed during run-up; MW or inlet pressure once synchronised
Evaporator pressure	Evaporator pressure and superheat (2)	Feed-pump VFD (10)	Vapour conditions at turbine inlet as heat input varies
Condensing pressure	Condenser pressure (9)	ACC fan VFDs (9)	Back-pressure near optimum across ambient swings
Voltage / VAR	Generator terminal voltage and current (6)	Exciter field current (6)	Terminal voltage, or the power-factor / MVAR setpoint
Synchronisation	Voltage, frequency and phase across 52G	Governor and AVR bias, 52G close coil (8)	Match across the breaker before closing is permitted
Plant / POI	MW, MVAR and voltage at the POI meter	Setpoints to all controllers above	Dispatch target and the POI voltage / power-factor schedule

Table 2: Control loops of the ORC power block. Numbers in parentheses refer to Table 1.

condensing pressure, feed pump and fans — are far too slow to influence stability timeframes and disappear into the boundary condition $P_{\text{mech}} \approx \text{const.}$ What remains is the wiring of Figure 2, which is identical in every tool; only the block names change.

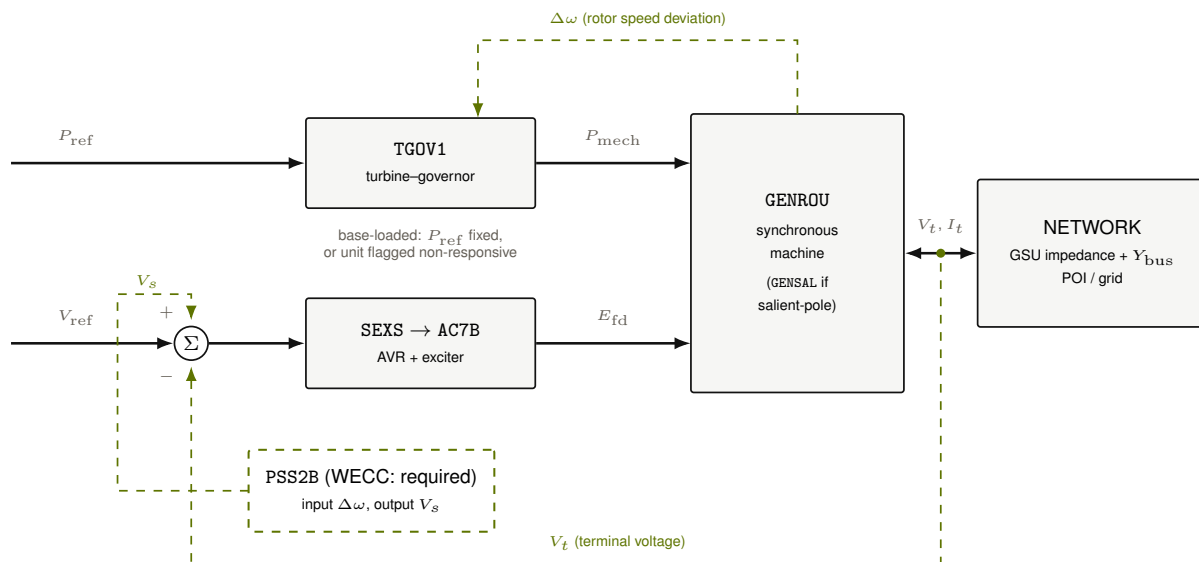


Figure 2: Model wiring implemented by every positive-sequence tool. Solid lines are model signals; dashed lines are the two feedbacks that close the loops — $\Delta\omega$ to the governor and V_t to the AVR summing junction. Protection and the synchroniser do not appear as dynamics.

1.5 Turbine–governor model

The governor block of Figure 2 is expanded in Figure 3. TGOV1 is the model named by the NERC turbine-governor modelling guideline for a simple, non-reheat thermal unit and is the appropriate choice for an ORC turbine.

1.6 Excitation system model

The excitation block of Figure 2 is expanded in Figure 4, shown here in its screening-level SEXS form.

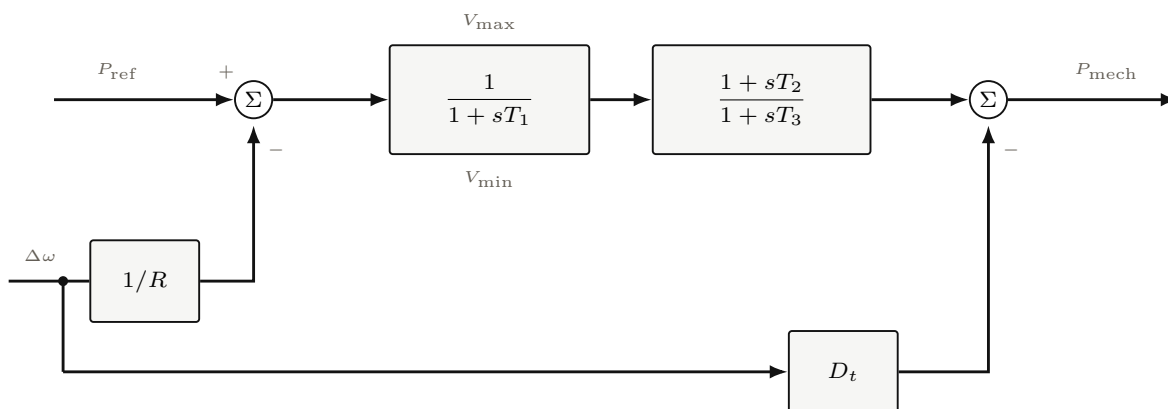


Figure 3: TGOV1 turbine-governor model. For an ORC unit T_1 lumps the valve stroke and the evaporator vapour-volume lag ($\approx 0.4\text{--}1.5\text{ s}$); there is no reheater, so $T_2 = T_3$ and the lead-lag drops out. Base-loaded operation is represented either by $V_{\max} = V_{\min} = P_0$ or by flagging the unit non-responsive.

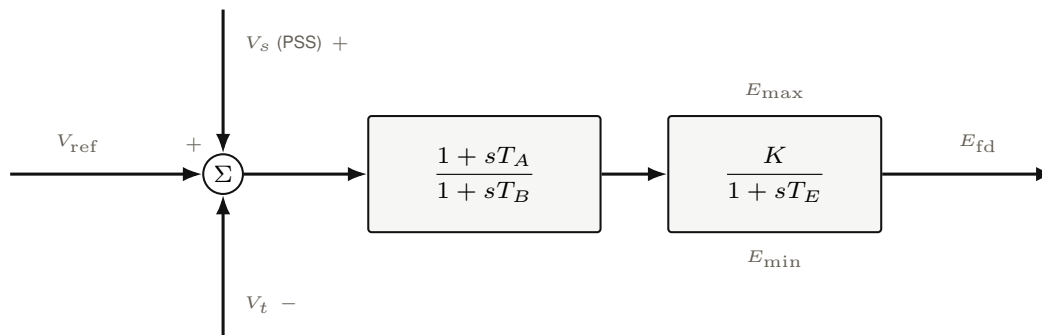


Figure 4: SEXS excitation model: voltage error, regulator lead-lag, exciter gain with ceiling limits. It is a screening-level placeholder; once OEM excitation data is available the block is replaced by IEEE AC7B/AC8B for a brushless exciter (the usual case on ORC machines) or ST4B for a static exciter. The wiring of Figure 2 is unchanged.

1.7 Power system stabilizer and WECC requirement

A power system stabilizer (PSS) is a supplementary controller added to the excitation system — the dashed block in Figure 2. The AVR alone regulates terminal voltage, but a fast, high-gain AVR erodes the damping of the rotor's electromechanical oscillations. Following a disturbance every synchronous rotor swings against the rest of the system at 0.1–3 Hz: local machine-against-system modes around 1–2 Hz, and inter-area modes, in which large groups of machines in one region swing against another, at 0.1–0.8 Hz. The PSS measures rotor speed deviation and electrical power — the modern dual-input integral-of-accelerating-power design, models PSS2B and PSS2C — phase-compensates that signal and injects it as V_s into the AVR summing junction, so that the field is modulated in phase with rotor speed and produces a damping torque. The PSS adds damping; it does not alter steady-state voltage control.

The Western Interconnection has long transmission corridors and well-known inter-area modes — the ~0.25 Hz North–South mode was central to the August 1996 west-wide separation — and WECC is consequently the one region with its own enforceable regional PSS standard.

VAR-501-WECC-4 requirements. The standard applies to synchronous generators connected to the Bulk Electric System that meet the definition of Commercial Operation; a 40 MW / 47 MVA unit falls within scope. The obligations relevant to this project are:

- **Installation** — PSS installed and placed in service within 180 days of a new generator connecting to the BES, or of the replacement of an existing voltage regulator.
- **In service** — the PSS must be in service whenever the unit is synchronised, other than during component failure, testing or maintenance, or as otherwise agreed with the Transmission Operator. Outages shorter than 30 minutes are not violations; inoperable equipment must be repaired or replaced within 24 months.
- **Tuning** — phase response within $\pm 30^\circ$ over the defined frequency range; output authority of at least $\pm 5\%$ of terminal voltage; gain set between one-third and one-half of the maximum practical gain; washout time constant not exceeding 30 s.
- **Exemption** — narrow, and limited to excitation systems incapable of meeting the tuning requirements; it lapses when the voltage regulator is replaced or retrofitted.

The practical consequences for this project are that the PSS must be specified together with the excitation system at the time of purchase, that commissioning must include PSS tuning tests, and that a PSS2B/PSS2C record must accompany the exciter model in the submitted dynamics data, populated from those tests.

1.8 Model parameters

Table 3 gives screening-level values on a 47 MVA machine base (40 MW at 0.85 power factor), 60 Hz, four-pole through gearbox. The characteristically ORC parameter is the **low inertia**: a compact expander and gearbox store far less rotational energy than a steam turbine train, so H sits at the bottom of the synchronous range.

The corresponding PSS/E dynamics record is:

```
/ 40 MW ORC unit at bus 101, id 1 - machine base 47.0 MVA, 60 Hz
/ Placeholder screening data: replace with OEM values for any filing
101 'GENROU' 1   5.00  0.05  0.80  0.08  1.80  0.00
                1.80  1.70  0.30  0.55  0.20  0.12  0.10  0.40 /
101 'SEXS'   1   0.10  10.0  100.0  0.10  0.00  4.00 /
101 'TGOV1'  1   0.05  0.40  1.00  0.00  1.00  1.00  0.00 /
```

A PSS2B record must be added alongside the exciter before submission, per Section 1.7.

Model	Typical parameters
GENROU	$T'_{d0} = 5.0 \text{ s}$, $T''_{d0} = 0.05$, $T'_{q0} = 0.8$, $T''_{q0} = 0.08$, $H = 1.8 \text{ s}$, $D = 0$, $X_d = 1.80$, $X_q = 1.70$, $X'_d = 0.30$, $X'_q = 0.55$, $X'' = 0.20$, $X_l = 0.12$, $S(1.0) = 0.10$, $S(1.2) = 0.40$. $H \approx 1.5\text{--}2.5 \text{ s}$ is typical for ORC turbogenerators. Use GENSAL/GENTPJ if the OEM machine is salient-pole, which is common at 1,500/1,800 rpm.
SEXS	$T_A/T_B = 0.10$, $T_B = 10.0 \text{ s}$, $K = 100$, $T_E = 0.10 \text{ s}$, $E_{\min} = 0.0$, $E_{\max} = 4.0$. Placeholder until the OEM AVR data sheet supports AC7B or ST4B.
TGOV1	$R = 0.05$, $T_1 = 0.40 \text{ s}$, $V_{\max} = 1.0$, $V_{\min} = 0.0$, $T_2 = 1.0$, $T_3 = 1.0$, $D_t = 0$. For a base-loaded unit set $V_{\max} = V_{\min} = P_0$, or omit the governor and flag the unit non-responsive.

Table 3: Screening-level dynamic model parameters, 40 MW ORC unit on a 47 MVA base. Replace with OEM data before any interconnection filing.

1.9 Portability across tools

The same three blocks exist under the same or near-identical names in every environment:

- **PSLF** — genrou / sexs / tgov1, with an epcl equivalent of the .dyr record.
- **PowerFactory, ETAP** — shipped as standard IEEE model implementations; parameters transfer directly from Table 3.
- **MATLAB/Simulink** — Figures 3 and 4 are wired directly from transfer-function blocks around the built-in synchronous machine model.
- **PSCAD (EMT)** — retain the machine and exciter and drive the shaft with constant mechanical torque. The ORC thermal side contributes nothing at EMT timescales.

1.10 Open items

- Obtain the OEM generator data sheet and confirm round-rotor versus salient-pole construction, which selects GENROU against GENSAL/GENTPJ.
- Obtain the excitation system type and data to replace SEXS with AC7B/AC8B or ST4B.
- Confirm whether the unit will provide primary frequency response; if not, agree the non-responsive representation with the Transmission Planner.
- Specify the PSS with the excitation system and schedule commissioning tuning tests (Section 1.7).
- Obtain GSU impedance, machine saturation data and the reactive capability curve. These bind the model to the real plant more tightly than any control parameter.